

Programming assignment 12

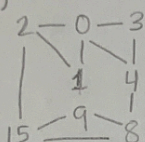
1 Undirected graphs

1-1)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	0	0	0	3	13	7	6	4	1	6	10	13	5	5	1
2	2	1	4	8	14	10	10	9	8	7	14	12	12	2	
3	9	15		0				15	15	11		14	13	8	
4	15													9	

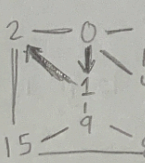
1-2)

1. DFS(0)



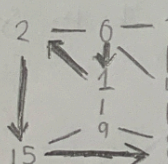
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

3. DFS(2)



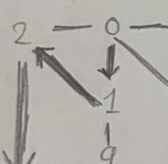
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

5. DFS(8)



0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

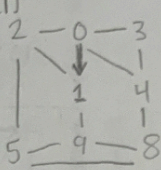
7. DFS(3)



0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

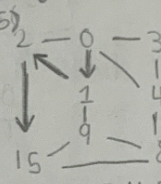
2. DFS(1)

2. DFS(1)



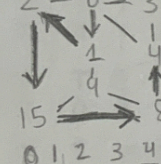
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

4. DFS(15)



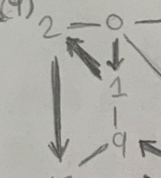
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

6. DFS(4)



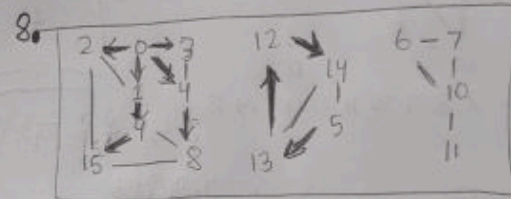
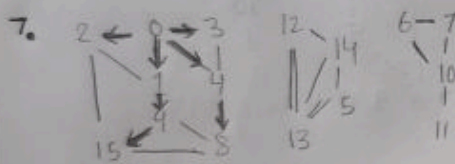
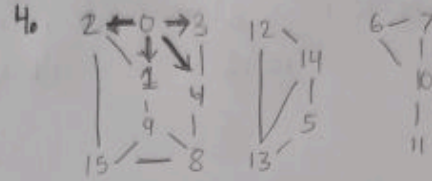
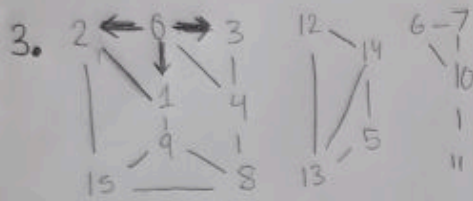
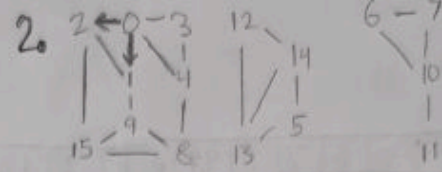
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

8. DFS(9)



0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

1-3)

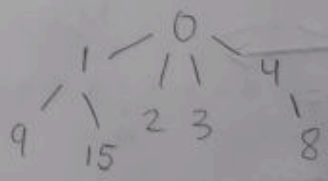


final:

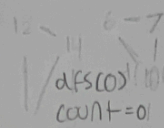
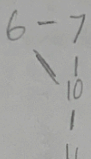
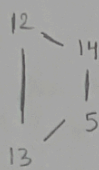
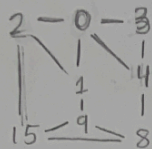
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
T	T	T	T	T				T	T						T
	0	0	0	0				9	1						1

Queue: 0 1 2 3 4 9 15 8

tree built:



1-4)

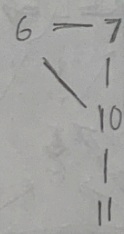
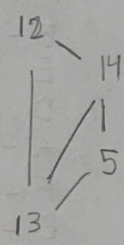
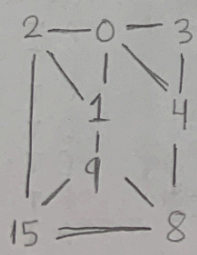


dfs(0) marked →
count = 0 ID →

dfs(1) c=0
dfs(2) c=0

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
dfs(0)	T															
dfs(1)	T	T														
dfs(2)	T	T	T													
dfs(3)	T	T	T	T												
dfs(4)	T	T	T	T	T											
dfs(5)	T	T	T	T	T	T										
dfs(6)	T	T	T	T	T	T	T									
dfs(7)	T	T	T	T	T	T	T	T								
dfs(8)	T	T	T	T	T	T	T	T	T							
dfs(9)	T	T	T	T	T	T	T	T	T	T						
dfs(10)	T	T	T	T	T	T	T	T	T	T	T					
dfs(11)	T	T	T	T	T	T	T	T	T	T	T	T				
dfs(12)	T	T	T	T	T	T	T	T	T	T	T	T	T			
dfs(13)	T	T	T	T	T	T	T	T	T	T	T	T	T	T		
dfs(14)	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	
dfs(15)	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T

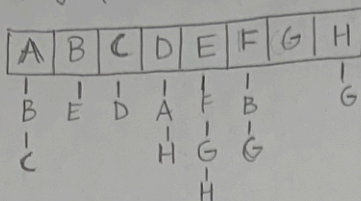
1-5)



dfs(0,0) T
dfs(1,0) T T
dfs(2,1) T T
Stops here

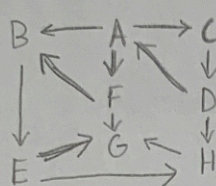
2) Directed graphs

2-1)



2-2)

1. dfs(A)



adjacent

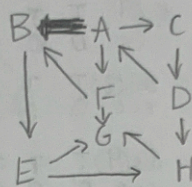
B-C

E
D
A-H
F-G-H
B-G
H

marked

A T
B T
C
D
E
F
G
H

2. dfs(B)



adjacent

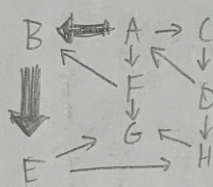
B-C

E
D
A-H
F-G-H
B-G
H

marked

A T
B T
C
D
E
F
G
H

3. dfs(E)



adjacent

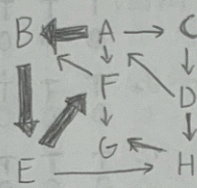
B-C

E
D
A-H
F-G-H
B-G
H

marked

A T
B T
C
D
E T
F
G
H

4. dfs(F)



adjacent

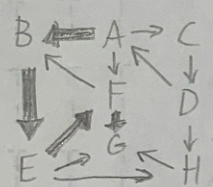
B-C

E
D
A-H
F-G-H
B-G
H

marked

A T
B T
C
D
E T
F T
G
H

5. dfs(G)



adjacent

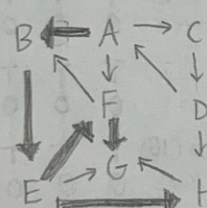
B-C

E
D
A-H
F-G-H
B-G
H

marked

A T
B T
C
D
E T
F T
G T
H

6. dfs(H)



adjacent

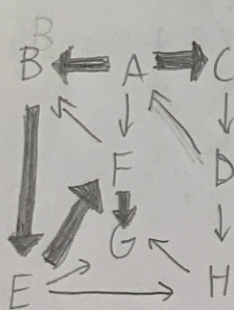
B-C

E
D
A-H
F-G-H
B-G
H

marked

A T
B T
C
D
E T
F T
G T
H T

7. dfs(C)



adjacent

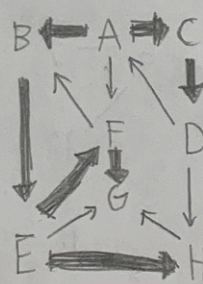
B-C

E
D
A-H
F-G-H
B-G
H

marked

A T
B T
C T
D
E T
F T
G T
H T

8. dfs(D)



adjacent

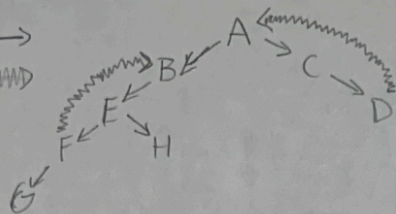
B-C

E
D
A-H
F-G-H
B-G
H

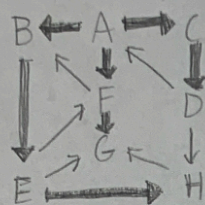
marked

A T
B T
C T
D T
E T
F T
G T
H T

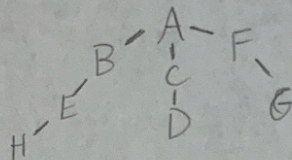
So tree edge: $\rightarrow$
back edge: $\curvearrowright$



2-3)

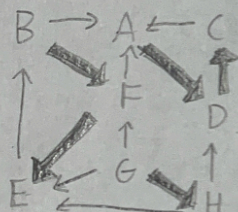


queue: A B C F E D G H



2-4)

1. G^R :



2. dfs for an unmarked V on G^R , going A-Z:

dfs(A)
dfs(D)
dfs(C) $\rightarrow$ "C"
D
A

dfs(B)
dfs(F)
dfs(E) $\rightarrow$ E
F
B

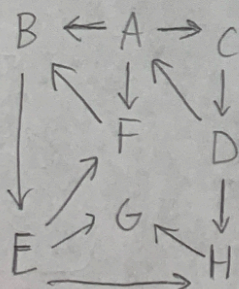
dfs(G)
dfs(H) $\rightarrow$ H
G

final:

C D A E F B H G

marked: A B C D E F G
T T T T T T T

3. dfs in order of reversed stack, so GHBFEDCA for G^{nr}



dfs(G) $\rightarrow$ G
dfs(H) $\rightarrow$ H
dfs(B) $\rightarrow$
dfs(E) $\rightarrow$ FEB
dfs(A) $\rightarrow$
dfs(C) $\rightarrow$
dfs(D) $\rightarrow$ DCA

G
H

F
E
B

D
C
A

strongly-connected components

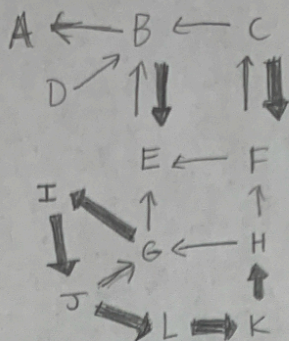
{G} {H}

{B E F}

{A C D}

3-1) $A \rightarrow B \rightarrow C$
 $\quad \quad \quad \swarrow \quad \updownarrow \quad \updownarrow$
 $\quad \quad \quad D \quad E \rightarrow F$
 $\quad \quad \quad \quad \downarrow \quad \downarrow$
 $I \rightarrow G \rightarrow H$
 $\quad \uparrow \quad \swarrow \quad \downarrow$
 $J \leftarrow L \leftarrow K$

→

[illegible]
$$dfs(B)$$
$$d\psi(E) = E$$
B
A
$$dfs(c)$$
$$df_s(F) = F$$
$$\text{dfs}(D) = D$$
$$\text{dfs}(G)$$
$$df_s(I)$$
$$df_s(J)$$
$$df_S(L)$$
$$df_3(K)$$
$$df_5(H)$$

1 H
K
L
J
I
G

So our stack is

G I J L K H D C F B E A

marked: G I J L K H D C F B E A
T T T T T T T T

$$\text{dfs}(G)$$

dfs (J)

$$d\beta(I) - I$$
$$\text{dfs}(D) = D$$
$$dfs(C)$$
$$dfs(F) - E$$

dfs(H)

des (K)

$$\text{dfs}(L) = L$$

K
H
G

$$dfs(B)$$
$$\Delta F_s(E) - E$$
$$df_s(A) \quad B$$
$$df_s(A) - A$$

I ↑
J ↑
L ↑
K ↑
H ↑
G
D
F ↑
C

$$E \downarrow B \quad A$$

Strongly
connected components
are:

$$\{G J I H K L\}$$
$$\{D\} \{CF\}$$
$$\{B, E\} \quad \{A\}$$